

$$e_1, e_2 ::= \underline{n} \mid \text{true} \mid \text{false} \mid \dots \mid x \mid e.attr \mid e_1[e_2] \mid e_1(\vec{e_2}) \mid [e_1, \dots, e_n] \mid \{e_1 : e'_1, \dots, e_n : e'_n\} \mid \text{new}$$

$$s ::= x = e \mid x.attr = e \mid x[e_1] = e_2 \mid \vec{s} \mid \text{if } e : s_1 \text{ else } : s_2 \mid \text{return } e \mid \text{def } f(\vec{x}) : s$$

$$A, B ::= \text{int} \mid \text{bool} \mid o_i \mid \text{List} \mid \text{Dict} \mid \text{Any} \mid \vec{A} \rightarrow B$$

VAR	NUM	...	DICT
$\Theta; \Gamma, x : A \vdash x : A$	$\Theta; \Gamma \vdash \underline{n} : \text{int}$		$\frac{(\forall i \in [1, n]) \Theta; \Gamma \vdash e_i : \text{Any} \quad \Theta; \Gamma \vdash e'_i : \text{Any}}{\Theta; \Gamma \vdash \{e_1 : e'_1, \dots, e_n : e'_n\} : \text{Dict}}$
LIST	NEW		IND-1
$\frac{(\forall i \in [1, n]) \Theta; \Gamma \vdash e_i : \text{Any}}{\Theta; \Gamma \vdash [e_1, \dots, e_n] : \text{List}}$	$\frac{}{\Theta; \Gamma \vdash \text{new} : o_i} \text{ } o_i \text{ is fresh}$		$\frac{\Theta; \Gamma \vdash e_1 : \text{Dict}}{\Theta; \Gamma \vdash e_1[e_2] : \text{Any}}$
IND-2	ATTR		
$\frac{\Theta; \Gamma \vdash e_1 : \text{List} \quad \Theta; \Gamma \vdash e_2 : \text{int}}{\Theta; \Gamma \vdash e_1[e_2] : \text{Any}}$	$\frac{\Theta, o_i.attr : A; \Gamma \vdash e : o_i}{\Theta, o_i.attr : A; \Gamma \vdash e.attr : A}$		
ASSIGN-VAR	ASSIGN-ATTR		RETURN
$\frac{\Theta; \Gamma \vdash e : A}{\Theta; \Gamma \Vdash x = e : \text{empty} \ \& \ x \mapsto A}$	$\frac{\Theta; \Gamma \vdash e : A}{\Theta; \Gamma, x : o_i \Vdash x.attr = e : \text{empty} \ \& \ o_i.attr \mapsto A}$		$\frac{\Theta; \Gamma \vdash e : A}{\Theta; \Gamma \Vdash \text{return } e : A \ \& \ \varepsilon}$
IF	SEQ		FUN
$\frac{\Theta; \Gamma \vdash e : \text{bool} \quad \Theta; \Gamma \Vdash s_1 : A \ \& \ \psi_1 \quad \Theta; \Gamma \Vdash s_2 : A \ \& \ \psi_2}{\Theta; \Gamma \Vdash \text{if } e : s_1 \text{ else } : s_2 : A \ \& \ \psi_1 \vee \psi_2}$	$\frac{\dots}{\dots}$		$\frac{\dots}{\dots}$